AGENTIC AI ENGINEERING

Assignment 1 - Intelligent Student Support Agent

Build one working agent and show that it can use tools, memory and relevant context effectively.

PGDM (Big Data Analytics / Artificial Intelligence & Data Science) | Batch 2025-27

Assignment Description

Build an AI Student Support Agent that can answer common student queries such as fees, scholarships, course changes and support requests. The system should demonstrate the core agentic concepts covered in the course through one working implementation.

Expected final output: one chat-style interface where a student can ask questions and the same agent can decide when to answer directly, use a tool, retrieve policy information, or use remembered information.

What the Agent Should Demonstrate

- A working agent loop that can decide when a tool is required and use the tool result before answering.
- At least two useful tools, for example checking a student record, fee status, scholarship status or creating a support request.
- Persistent memory across sessions for useful information such as programme or preferences, while temporary or sensitive information should not be stored.
- Retrieval from one student-policy / handbook document. The system should retrieve only relevant information rather than sending the whole document every time.
- Context selection: the model should receive the current question plus only the information needed for that question.
- Basic handling of missing information, tool errors or a question that cannot be answered from the available data.

Tracing and Measurement

For a defined set of test questions, show a trace for each run. A table or structured log may be used to present the trace.

Measure	What to show
Task accuracy / success	How many test questions were answered correctly or completed successfully.
Tools used	Which tool was called and whether the tool choice was appropriate.
Latency	Approximate end-to-end response time for each test run.
Token usage	Input and output token usage reported by the model API where available.
Retrieved evidence	Which policy chunk(s) or memory items were used for the answer.

Suggested Demonstration Scenarios

- “What is my current fee status?”
- “The payment portal failed. Can I pay on the next working day?”
- A new session where the student asks about a useful fact discussed earlier.
- A temporary or sensitive statement that should not be stored as long-term memory.
- A question for which the available policy does not contain enough information.

Submission

- Working implementation of the Student Support Agent.
- The sample data / policy document used by the system.
- A short README explaining the architecture and how to run the agent.
- An evaluation table showing accuracy, latency, token usage and key observations.

Keep the solution focused on the required agentic concepts and ensure that each implemented capability can be clearly explained during the demonstration.